UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER II, SESSION 2024/2025

SUBJECT CODE **: SECI 1113**

SUBJECT NAME **: COMPUTATIONAL MATHEMATICS**

SECTION **: 01/02/03/04**

TIME **:**

DATE/DAY **:**

VENUES **:**

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

INSTRUCTIONS:

ANSWER ALL QUESTIONS IN THE ANSWER BOOKLET

(Write Your Name and Section in Your Answer Booklet)

Name	
IC/ISID No.	
Year / Course	
Section	
Lecturer Name	

WARNING: Disciplinary action will be taken against students who are found cheating during the examination.

This question paper consists of **SIX (6)** printed pages including this page.

QUESTION 1 **[20 MARKS]**

a) Given $f(x) = 4x^3 - 2x^2 + 1$

i) Find the exact value of $f(2.13)$. (1 mark)

ii) Find the absolute and relative error if $f(2.13)$ is round up to 3 decimal points. (3 marks)

b) Find the root of the equation below with accuracy of 1 decimal point in the interval [1, 3] using bisection method. Do calculation in 3 decimal points. (8 marks)

$$f(x) = x^3 - 4x + 1$$

c) Find the root with accuracy to 2 decimal points for the equation below using Secant method, where $x_0 = 1$ and $x_1 = 1.5$. Do calculation in 4 decimal points. (8 Marks)

$$f(x) = cox(x) - x$$

QUESTION 2 **[20 MARKS]**

a) Given matrix *A*,

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

i) Use characteristic polynomial to find the eigenvalue and eigenvector for *A*. (5 marks)

ii) Show that eigen values obtain in (i) fulfill the theorem $\sum_{i=1}^{2} \lambda_i = \sum_{i=1}^{2} a_{ii}$ (1 mark)

b) Given matrix *B*,

$$B = \begin{bmatrix} 4 & 1 & -1 \\ 2 & 3 & 1 \\ -1 & 1 & 2 \end{bmatrix}$$

i) Determine the smallest in magnitude the eigenvalue λ_3 of B and its associated eigenvector, v_3. Use shifted power method and suppose the largest magnitude of eigenvalue for matrix *B*, $\lambda_1 = 5$. Start your iteration with $v_0 = [0.5\ -0.4\ 0.5]^T$ and iterate until $\mathcal{E} = 0.05$. (12 mark)

ii) Find the intermediate eigenvalue. (2 marks)

QUESTION 3 **[20 MARKS]**

An object is moving with a non-uniform speed. The position of the object at specific time interval are recorded as in Table 1.

Table 1

Time (s)	1	2	3	4	5
Position (m)	2.0	3.5	7.8	13.9	21.0

a) Estimate the position of the object at *t=4.4* seconds using: -

i) appropriate Newton method` (8 Marks)

ii) Lagrange interpolating polynomial. Given: $L_0 = -0.03$. $L_1 = -0.19$ and $L_2 = -0.49$ (5 Marks)

b) Determine the appropriate polynomial expression, $p(x) = a_0 + a_1 x$, using Least Squares Approximation. (7 Marks)

Note: Do all calculations to 2 decimal points

QUESTION 4 **[20 MARKS]**

Table 2 shows values of a function $f(x) = \ln(x^2 + 1)$ measured at specific data points:

Table 2

x	$f(x) = \ln(x^2 + 1)$
0.9	0.58779
1.0	0.69315
1.1	0.78846
1.2	0.87547
1.3	0.95551
1.4	1.02962

a) Estimate the **first derivative** $f'(1.0)$ using $h = 0.1$ with the following numerical differentiation formulas:

i) Two-point forward difference. (2 marks)

ii) Three-point forward difference. (4 marks)

iii) Five-point forward formula. (6 marks)

Note: You may need to compute additional values for insufficient data point.

b) Given the analytically derivative of $f(x) = \ln(x^2+1)$ is $f'(x) = \frac{2x}{x^2+1}$. Calculate the *absolute error* for each derivative estimated in part (a) by comparing with exact value obtained using the analytical derivative. What do you observe? Explain how using more data points affect the accuracy approximation values. (3 marks)

c) Estimate the **second derivative** $f''(1.1)$ with $h = 0.1$ using five-point central difference formula. (5 marks)

QUESTION 5 **[20 MARKS]**

Approximate the integral $\int_1^4 \left(\frac{x^2-x+1}{x^3}\right)dx$ using the following methods:

a) Trapezoidal approximation, $N = 6$. (4 marks)

b) Simpson's 1/3 approximation, $N = 6$. (4 marks)

c) Romberg integration. Calculate until $R_{3,3}$. (12 marks)

-End Question -

List of Formula:

Non linear equation

Bisection Method, $x_c = \frac{a+b}{2}$

Secant Method $x_{i+2} = \frac{x_i f(x_{i+1}) - x_{i+1} f(x_i)}{f(x_{i+1}) - f(x_i)}$, *where i=0,12,…n*

Eigenvalue

$v^{(k+1)} = \frac{1}{m_{k+1}} A_{shifted} v^{(k)}$, *where k=0,12,…*

Interpolation and Approximation

Newton Forward Difference Formula:

$$p_n(x) = y_k + r\Delta y_k + \frac{r(r-1)}{2!}\Delta^2 y_k + \ldots + \frac{r(r-1)\ldots(r-n+1)}{n!}\Delta^n y_k$$

with $r = (x - x_k)/h$.

Newton Backward Difference Formula:

$$p_n(x) = y_k + r\nabla y_k + \frac{r(r+1)}{2!}\nabla^2 y_k + \ldots + \frac{r(r+1)\ldots(r+n-1)}{n!}\nabla^n y_k$$

with $r = (x - x_k)/h$.

Lagrange Formula:

$$p_n(x) = L_0(x)y_0 + L_1(x)y_1 + \ldots\ldots L_n(x)y_n = \sum_{i=0}^{n} L_i(x)y_i$$

with

$$L_i(x) = \frac{(x-x_0)\ldots(x-x_{i-1})(x-x_{i+1})\ldots(x-x_n)}{(x_i-x_0)\ldots(x_i-x_{i-1})(x_i-x_{i+1})\ldots(x_i-x_n)} = \prod_{\substack{j=0 \\ j\neq i}}^{n} \frac{(x-x_j)}{(x_i-x_j)}$$

Numerical Differentiation

Three -Point Central Difference

$$f'(x_i) \approx \frac{f(x_i+h) - f(x_i-h)}{2h}$$

Five-Point Forward Difference

$$f'(x_i) \approx \frac{1}{12h}[-25f(x_i) + 48f(x_i+h) - 36f(x_i+2h) + 16f(x_i+3h) - 3f(x_i+4h)]$$

Five-Point Central Difference

$$f'(x_i) \approx \frac{1}{12h}[f(x_i - 2h) - 8f(x_i - h) + 8f(x_i + h) - f(x_i + 2h)]$$

$$f''(x_i) \approx \frac{1}{12h^2}[-f(x_i - 2h) + 16f(x_i - h) - 30f(x_i) + 16f(x_i + h) - f(x_i + 2h)]$$

<u>Numerical Integration</u>

Trapezoidal Rule

$$\int_a^b f(x) \quad dx = \frac{h}{2}\left(f_0 + f_N + 2\sum_{i=1}^{N-1} f_i \right)$$

Simpson

$$\int_a^b f(x) \quad dx = \frac{h}{3}\left[(f_0 + f_N) + 4\sum_{i=1}^{N/2} f_{2i-1} + 2\sum_{i=1}^{N/2-1} f_{2i} \right]$$

$$\int_a^b f(x) \quad dx = \frac{3h}{8}\left[(f_0 + f_N) + 3\sum_{i=1}^{N/3} (f_{3i-2} + f_{3i-1}) + 2\sum_{i=1}^{N/3-1} f_{3i} \right]$$

Romberg Integration

$$h_i = \frac{1}{2} h_{i-1}$$

$$R_{1,1} = \frac{h_1}{2}(f_0 + f_1)$$

$$R_{i,1} = \frac{1}{2}\left[R_{i-1,1} + h_{i-1}\sum_{k=1}^{2^{i-2}} f_{2k-1} \right] \text{for } i = 1,2,3....$$

$$R_{i,j} = \frac{4^{j-1} R_{i,j-1} - R_{i-1,j-1}}{4^{j-1} - 1} \text{ for } i = 2,3...N \text{ and } j = 2,3,..i$$

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